

Lecture 6: Program Execution & Interrupts (MCQs)

1. An event external to the current executing process that causes a change in the normal flow of instruction execution is called a(n) _____. a) Loop b) Interrupt c) Operand d) Fetch

• **Answer:** b) Interrupt

2. Internal interrupts, also known as _____, are caused by illegal operations or erroneous use of data. a) Traps b) External Interrupts c) Software Interrupts d) Vectors

• **Answer:** a) Traps,

3. Which of the following is an example of an Internal Interrupt (Trap)? a) I/O device request b) Division by zero c) Timer tick d) Keyboard press

• **Answer:** b) Division by zero

4. _____ interrupts occur specifically during the execution of an instruction, such as switching from user mode to supervisor mode. a) External b) Hardware c) Software d) Daisy Chain

• **Answer:** c) Software

5. In the hypothetical machine described in the lecture, the instruction 1940 (Hex) implies: Opcode 1 and Address 940. Opcode 1 represents _____. a) STORE to Memory b) ADD to AC c) LOAD AC from Memory d) Jump

• **Answer:** c) LOAD AC from Memory

6. In the same hypothetical machine, the opcode 5 (e.g., 5941) represents which operation? a) LOAD b) STORE c) ADD d) SUB

• **Answer:** c) ADD

7. When the processor detects an interrupt, the first step it takes is to _____. a) Immediately jump to the handler b) Finish the execution of the current instruction c) Reset the PC d) Clear the Accumulator

• **Answer:** b) Finish the execution of the current instruction

8. After acknowledging an interrupt, the processor must _____ to allow resuming the program later. a) Save the state (including PC) b) Delete the current instruction c) Turn off the I/O device d) Execute the next instruction

• **Answer:** a) Save the state (including PC)

9. The address of the Interrupt Service Routine (ISR) is often found using a(n) _____. a) Interrupt Vector b) Opcode c) Stack Pointer d) Accumulator

• **Answer:** a) Interrupt Vector

10. In the method for handling multiple interrupts, the processor _____. a)
Checks a hardware priority line b) Polls each I/O module one by one to find the source c)
Uses a daisy chain d) Accepts all interrupts simultaneously

• **Answer:** b) Polls each I/O module one by one to find the source

11. Which method of handling multiple interrupts uses a common "Interrupt Acknowledge" line that propagates through devices in series? a) Software Poll b)
Multiple Interrupt Lines c) Daisy Chain d) Bus Arbitration

• **Answer:** c) Daisy Chain

12. In a , the priority of a module is determined by _____. a) Software assignment
b) The position of the module in the chain c) The speed of the device d) Random
selection

• **Answer:** b) The position of the module in the chain

13. In the hypothetical machine, if the instruction is 2941, and Opcode 2 is STORE,
what happens? a) Content of address 941 is loaded to AC b) Content of AC is stored
into address 941 c) Content of AC is added to address 941 d) The program halts

• **Answer:** b) Content of AC is stored into address 941

14. Which interrupt handling method is slow because it relies on jumping to a
routine that checks every device? a) Daisy Chain b) Hardware Polling c) Software Poll
d) Vectored Interrupt

• **Answer:** c) Software Poll

15. In , the _____ scheme uses a single hardware device (Arbiter) to allocate
time on the bus. a) Distributed b) Centralized c) Daisy Chain d) Parallel

• **Answer:** b) Centralized

16. In bus arbitration, _____. a) There is no central controller; modules act together
b) The CPU always controls the bus c) An Arbiter decides who goes next d) It is
impossible to handle multiple requests

• **Answer:** a) There is no central controller; modules act together

17. If multiple interrupts occur, and the processor has , it picks the line with the
_____. a) Lowest voltage b) Highest priority c) First signal d) Last signal

• **Answer:** b) Highest priority

18. The value 0011 in the hypothetical machine instruction set represents: Load AC
from ... a) Memory b) I/O c) Register d) Cache

- **Answer:** b) I/O

19. In the Daisy Chain method, the requesting module responds to an acknowledge signal by placing a _____ on the data lines. a) Status bit b) Vector (word) c) Flag d) Error code

- **Answer:** b) Vector (word)

20. External Interrupts generally come from _____. a) The ALU b) External input/output devices c) Illegal instructions d) Software commands

- **Answer:** b) External input/output devices,

21. When an interrupt handler finishes, the processor must _____. a) Halt the system b) Restore the saved register values and PC c) Restart the computer d) Poll the devices again

- **Answer:** b) Restore the saved register values and PC

22. Which of the following is NOT a cause of an Internal Interrupt (Trap)? a) Invalid Opcode b) Division by Zero c) Printer requesting paper d) Stack Fault

- **Answer:** c) Printer requesting paper (This is an External Interrupt)

23. In the instruction execution trace, the PC (Program Counter) holds the address of the _____ instruction. a) Previous b) Current c) Next d) Last executed

- **Answer:** c) Next

24. Which register holds the instruction currently being executed (to be decoded)? a) PC b) IR c) AC d) MAR

- **Answer:** b) IR

25. If the Daisy Chain signal propagates to the end without being intercepted, it means _____. a) The system is broken b) No device requested an interrupt c) The last device has the highest priority d) An error occurred

- **Answer:** b) No device requested an interrupt

26. 0111 in the hypothetical machine instructions represents _____. a) Store AC to I/O b) Load AC from I/O c) Add Memory to AC d) Halt

- **Answer:** a) Store AC to I/O

27. What happens if an interrupt occurs while the processor is in the middle of executing an instruction? a) It stops immediately and services the interrupt b) It discards the instruction c) It completes the instruction first d) It ignores the interrupt

- **Answer:** c) It completes the instruction first

28. In , identifying devices and setting priorities is necessary when _____. a)

Only the CPU uses the bus b) More than one module may need control of the bus c) The bus is dedicated d) The memory is full

• **Answer:** b) More than one module may need control of the bus

29. Interrupts allow the processor to handle _____ efficiently without waiting in

loops. a) Arithmetic operations b) High-priority conditions/events c) Cache misses d) Pipelining

• **Answer:** b) High-priority conditions/events

30. Which of the following is true about ? a) They are initiated by the user pressing a button. b) They are caused by the program itself (e.g., exceptions). c) They are used for normal I/O operations. d) They are masked by the system.

• **Answer:** b) They are caused by the program itself (e.g., exceptions)